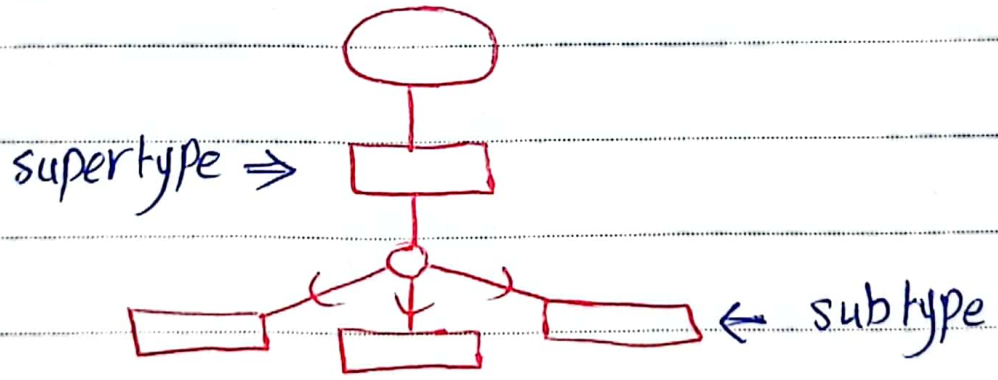
DATE / /

Chapter 3

※ enhanced entity ⇒ relationship diagrams, or EERDs are specialized ER diagrams that can be extremely.

Incorporate

- supertype ⇒ represents a general entity type.
- subtype ⇒ specialized entities



ملاحظة : العنصر الخاص يرث جميع ما فيه الموجودة في العنصر العام.

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* subtype inherits attributes from supertypes. and can have additional attributes and involve in relationships own.

※ generalization:

- defines a more general entity type from specialized types.
- bottom-up process.
- Combines lower-level entities into a higher level.
- forms a more general entity type by taking common attributes (specialized entity types).

※ Specialization:

- divides high-level entities into lower levels.
- top-down process.
- forms more specialized entity types by taking unique attributes.
- difining one or more subtypes of the supertyp and forming subtype/supertype relationships.

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* Super Class

generic entity type has relationship one or more subtypes.

* Sub class

group of entities with unique attributes. inherits all attributes from super class.

Inheritance

- is important feature of generalization & specialization.
- allows lower-level entities to inherit attributes from higher-level entities.
- subtype entities inherit values of all attributes and instances of relationships from their supertype.

- every member of a subtype must also be a member of the supertype.
- member of the supertype may or may not belong to any of its subtypes.

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* Constraints on EERD

important business rules that apply super class

① based on Membership
(Disjointness Constraint) — ما يمكن عمل عضوية مع الآخرين

② based on Participation
(Completeness Constraint). — يمكن عمل عضوية مع الآخرين

Based on Membership

addresses whether an ~~ins~~ instance of a supertype may simultaneously be a member of 2 or more subtypes.

rules:

→ Disjoint rule or Disjoint sub classes

(d) ⇒ an entity instance of the supertype is a member of one subtype.

cannot simultaneously be a member of any other subtype.

لا يمكن ان يكون عضوًا في اي نوع فرعي آخر في نفس الوقت.

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→ Overlap rule or Overlapping sub classes

(O) ⇒ an entity instance can simultaneously be a member of two or more subtypes.

يمكن ان يكون عضوًا في نوعين او أكثر في نفس الوقت

Based on Participation

addresses the question of whether an instance of a supertype must also be a member of at least 1 subtype.

rules:

→ total specialization / complete participation

(||) ⇒ each entity instance of a supertype must be a member of some subtype in the relationships.

→ Partial specialization / ~~Partial~~ complete participation

(|) ⇒ an entity instance of a supertype is allowed not to belong to any subtype.

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* When to use supertype/subtype relationships.

- There are attributes unique to subtype.
- The subtype participate in a relationship unique to that subtype.

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